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M-010 MISCELLANEOUS HEAT EXCHANGERS

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M 010 MISCELLANEOUS HEAT EXCHANGERS

1. SCOPE OF WORK

The miscellaneous heat exchangers specified in **Part 4, Subsection 1.2 "Summary of Work"**, in **Section 3 Technical Specifications** or as required by the Bidder's design shall also be subject to the design, testing and quality assurance requirements of this specification as applicable.

Heat Exchangers concerned are among others:

- Two (2) x 100% Closed Cooling Water Heat Exchangers for each Closed Cooling Water System per Block. (requirements for Cathodic Protection if required to be observed)
- Oil, generator and other coolers for the Gas Turbine Generator set
- Oil, generator and other coolers for the Steam Turbine Generator set
- Boiler Feed Pump coolers
- Miscellaneous sample coolers
- Coolers for the Service and Instrument Air Compressors
- Natural gas heaters

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 "Codes and Standards"**.

The codes and standards of the following organizations extracted from **Part 4, Subsection 2.5 "Codes and Standards"** or international equivalent standards are specifically applicable to the design, manufacture and testing of the equipment included in this Specification.

ANSI	American national Standards Institute
ASTM	American Society of Testing and Materials.
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization

HEI	Heat Exchange Institute
IEC	International Electrotechnical Commission
JIS	Japan Industrial Standard
NEMA	National Electrical Manufacturers Association
TEMA	Tubular Exchanger Manufacturers Association

3. DESIGN AND OPERATING CONDITIONS

Bidder shall fill in the operating and design conditions for each heat exchanger furnish by him on the data sheets included in **Part 5, Schedule 7**.

Equipment coolers including lubricating coolers shall be designed for a water side temperature rise of no more than a normal 8°C.

For environmental consideration, see **Part 4, Subsection 2.16**.

3.1 Closed Cooling Water Heat Exchangers

The closed cooling water heat exchanger will be used to lower the temperature of the demineralized water used primarily for bearing and equipment cooling. Seawater pumped by the Circulating Water Pumps shall be used as the cooling medium. Heat exchanger shall be the shell and tube type or alternatively the plate type .

The heat exchangers shall be designed based on the following conditions:

3.1.1	Heat exchanger name.....	Closed Cooling Water H.E.
3.1.2	Number required.....	Two
3.1.3	Capacity of each H.E.....	100 %
3.1.4	For shell and tube type only	Tube side Fluid: Cooling water/Sea water
3.1.5	For shell and tube type only	Shell side Fluid - Demineralized water
3.1.6	Cleanliness factor.....	80 %
3.1.7	Cooling fluid Temperature range.....(°C)	See Part 4, Subsection 1.1
3.1.8	Temperature	See Part 4, Subsection 1.1

	Cooling water in-let.....(°C)	
3.1.9	Temperature Cooling water out-let.....(°C)	Determined by Bidder Maximum temp. rise = 7°C
3.1.10	Temperature Hot water inlet.....(°C)	Determined by Bidder
3.1.11	Temperature Hot water outlet.....(°C)	Determined by Bidder, Shall not exceed 40°C
3.1.12	Design temperature Plate type HE Tube side (for shell and tube HE)(°C)	65 (minimum)
3.1.13	Design temperature Shell side(for shell and tube HE)(°C)	65 (minimum)
3.1.14	Design pressure (bar)	Determined by Bidder
3.1.15	Design flow tube side.....(l/s)	Determined by Bidder
3.1.16	Tube size(for shell and tube HE)(mm)	24 (minimum)
3.1.17	Tube wall thickness.....(mm)	0,5 (minimum)
3.1.18	Maximum tube velocity(m/s)	2,15
3.1.19	Nozzle interface.....	Flanged per Part 4 Section 3, M-001 Piping.
3.1.20	Shell material.....	Carbon steel ASTM-285 C
3.1.21	Channels & Nozzles.....	Carbon steel with rubber lining
3.1.22	Tube material.....	Titanium
3.1.23	Tube sheet material.....	Titanium Clad
3.1.24	Type of heat exchanger.....	Straight- Tube
3.1.25	Governing code.....	ASTM Section VIII
3.1.26	Is cathodic protection required...	Yes for shell and tube type heat exchangers
3.1.27	Tube side design pressure Plate Type HE design pressure.....	Determined by Bidder. This pressure is provided by the Cooling water from the Circulating Water Pumps specified in Part 4 Section 3, M-206. Bidder shall be responsible for

		establishing the final design pressure based on the pump shut off head, with a 10% margin.
3.1.28	Closed cooling water design flow	
	Cooling water flow for the steam turbine generator set and its auxiliaries.....(l/s)	Bidder to determine
	Cooling Water flow for other users.....(l/s)	Bidder to determine based on clause 3.1.29 herein
	-Margin, % of above flows....	15 % (minimum)
3.1.29	Design Thermal Load	By Bidder.
3.1.29.1		The cooling capacity each the heat exchangers shall be sufficient to cool the lubrication and control oil system the generator of the steam turbine generator set and other systems in the combined cycle power plant as necessary. The gas turbine generator set will have its own independent cooling system.
3.1.30	Cooling water analysis	Refer to Part 4, Subsection 1.1
3.1.31	Closed Cooling Water analysis	Shall be treated demineralized water as specified in Part 4, Section 3, M-401

4. DETAILED REQUIREMENTS

4.1 General

Design of the heat exchangers shall be in accordance with ASME Section VIII and TEMA. The design shall incorporate the following features:

The heat exchangers shall be designed for maximum accessibility for inspection, cleaning and repair. Each shell and tube type heat exchanger shall be so constructed that the channel covers can be removed without disconnecting any piping. With covers removed, all tubes, tube sheets and channels shall be easily accessible for cleaning.

Provisions shall be made for expansion and contraction of all parts. Heat exchangers shall be designed so that there will be no flow-induced vibrations.

For shell and tube type heat exchangers, where flanged joints are used between shells, channels and channel covers, flanges shall be accurately machined and arranged for external bolting. Adequate means shall be provided for breaking the joints between channels and channel covers.

Special attention shall be given in the selection of gaskets for joints between shells, channels and channel covers. Gaskets shall be selected on the basis of TEMA Standards.

Heat exchanger nozzles shall be designed to withstand the forces and moments imposed on them as determined by the Bidder.

Heat exchangers shall be capable of handling the maximum, expected heat requirements, as specified, with 6% of the tubes plugged; i.e. the heat exchangers shall have a, minimum of 6% excess tubes.

In cases where the ASME code and TEMA standards do not apply, the design and construction shall be based on a minimum safety factor of five.

The heat exchangers shall be provided with vent and drain connections as required. All portions of the heat exchanger shall be drainable.

Packed head tube sheet design will not be acceptable.

Nameplates shall be located on the side of the heat exchangers where they will be visible from the mounting floor.

4.2 Channels

Tubes shall be expanded into tube sheets by a roller- type expander or other method not injurious to tubes.

Channels shall be designed to avoid erosion of internal parts. Tube entrances at inlet to each pass shall be belled except where weld design will not permit.

Water channel partition plates shall be designed to prevent interpass leakage.

Nozzles shall be sized to prevent excessive velocities. Water nozzle velocities shall be not exceed 3m/s.

Two quick opening handholes 500 mm diameter or more shall be provided on the channel on each heat exchanger (if required).

4.3 Tube-Support Plates

Tube-support plates shall be provided by Bidder and arranged so as not to interfere with proper distribution through exchangers or with free drainage. The support plate spacing and thickness shall conform to TEMA requirements.

Tube holes in support plates shall be drilled to allow a free sliding fit and holes shall be slightly chamfered on both sides of plates.

4.4 Supports

Bracket-type supports or saddles capable of safely carrying the entire weight of the heat exchanger with all spaces flooded shall be provided by Bidder.

4.5 Inlet and Outlet Connection

Terminals on heat exchangers for piping connections shall be per the requirements on the specified **Part 4 Section 3, M-001 Piping**.

All miscellaneous piping furnished with the heat exchangers shall be in

accordance on the specified **Part 4 Section 3, M-001 Piping**.

The nozzle construction and method of attaching to heat exchangers shall be in strict accordance with the ASME Code requirements.

Carbon steel nozzles for pipe connections shall be made of material whose specified minimum strength and allowable stress is equal to or greater than the vessel material to which they connect but in no case shall they be of quality less than required by SA-105, or SA-106, Grade B.

4.6 Shell

Each shell side connection shall be furnished with impingement baffles to prevent tube erosion.

Each heat exchanger shall be provided with two handholes in the shell. The Bidder will specify the locations of them on the first review of submittal drawings after Contract Award.

4.7 Relief Valves

Shell and channels shall be furnished with adequate relief valves.

Relief valves for channels shall be of steel.

Relief valve pressure setting shall not exceed the maximum design working pressure Bidder of the channel, in each case. The exact setting shall be verified with the Owner.

Relief valves shall be of the enclosed spring type.

Relief valve body and trim material shall be suitable for the service intended in each case.

4.8 Interior Cleaning

Prior to assembly all interior ferrous surfaces of the heat exchangers shall be thoroughly blast-cleaned using metal shot or metallic grit so as to leave clean metal surfaces exposed. After cleaning all grit and other debris shall be completely removed from the heaters. No rust-inhibiting agents or coatings shall be used on interior surfaces of the heat exchangers.

4.9 Painting and Protective Coating

Before shipment all parts of the equipment shall be thoroughly cleaned of all mill scale, rust, grease and other foreign matter and shall be painted in accordance with the requirement as in **Part 4 Section 3, S-420 and S-430 Cleanliness and Painting**.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 Performance Test

The Bidder shall run a performance test after equipment is put in service, to ascertain that it is performing according to the guarantee in every respect. Any deficiencies brought out by this test shall be promptly corrected by Bidder at no cost to Owner.

5.2 Shop Test

Shell and channels shall be hydrostatically tested to 1.3 times, their design

pressure in accordance with the requirements of ASME Section VIII, refer also to **Part 4 Section 3, M-014**.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality control shall conform to requirements of **Part 4, Subsection 2.7**.

7. ELECTRICAL SPECIFICATIONS

The Bidder shall furnish and install all electrical equipment and accessories required for Miscellaneous Heat Exchangers.

For general electrical requirements, see **Part 4 Section 3, E-001**.

Cathodic Protection System is specified in Part 4 Section 3, E-260.

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